

Chapter 25

The History of Life on Earth

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

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Overview: Lost Worlds

- Past organisms were very different from those now alive.
- The **fossil record** shows **macroevolutionary changes over large time scales** including
 - The emergence of terrestrial vertebrates
 - The origin of photosynthesis
 - Long-term impacts of mass extinctions.

MacroEvolution: Large Scale Changes Over Time



Concept 25.1: Conditions on early Earth made the **origin of life** possible

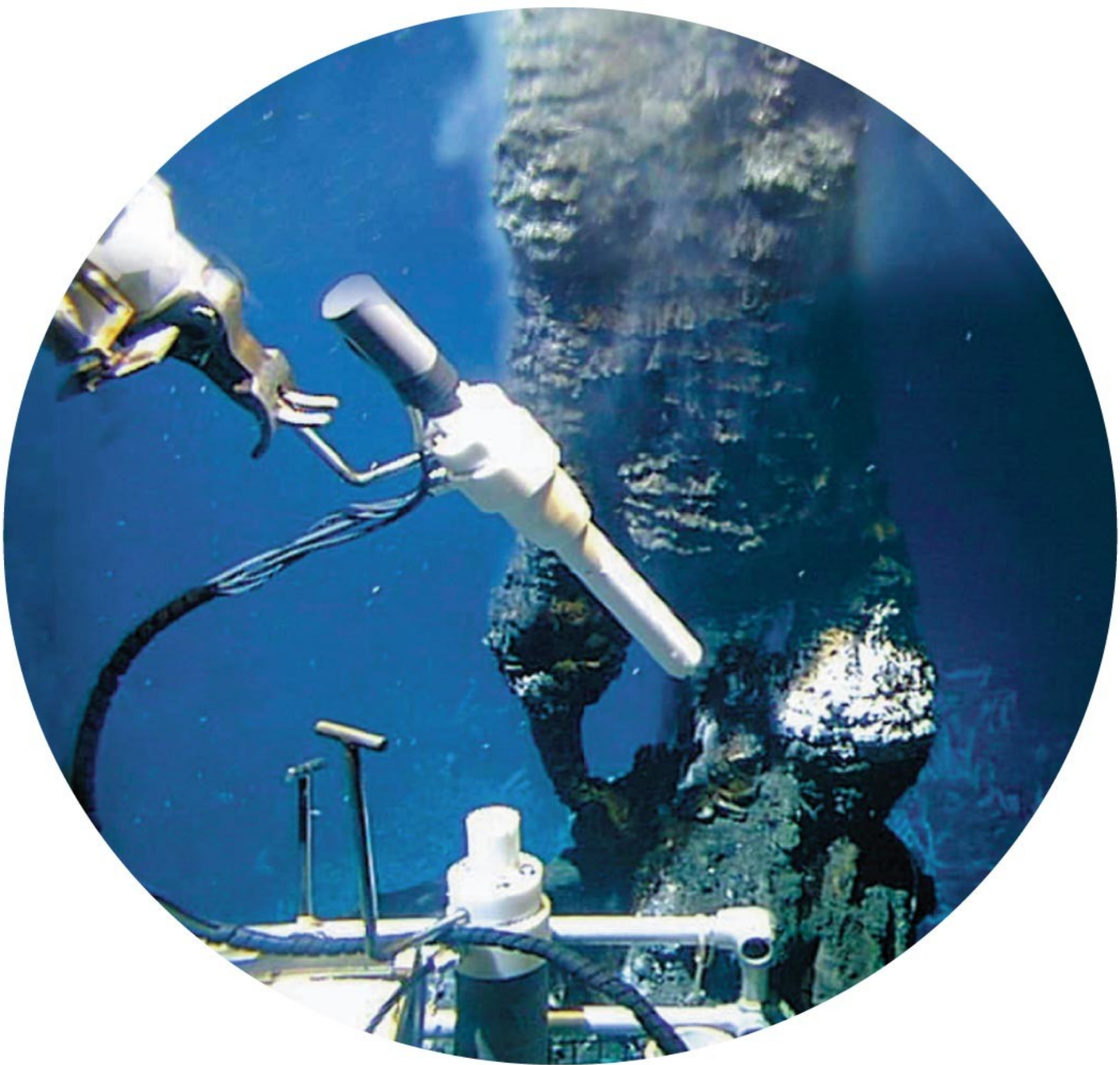
- Chemical and physical processes on early Earth may have produced very simple cells through a *sequence of stages*:
 1. Abiotic synthesis of small organic molecules.
 2. Joining of these small molecules into macromolecules.
 3. Packaging of molecules into “protobionts.”
 4. Origin of self-replicating molecules.

Synthesis of Organic Compounds on **Early Earth**

- Earth formed about 4.6 billion years ago, along with the rest of the solar system.
- Earth's early atmosphere likely contained water vapor and chemicals released by volcanic eruptions (nitrogen, nitrogen oxides, carbon dioxide, methane, ammonia, hydrogen, hydrogen sulfide).
- A. I. **Oparin** and J. B. S. **Haldane** hypothesized that the early atmosphere was ***a reducing environment***.
- **Stanley Miller** and Harold Urey conducted lab experiments that showed that the ***abiotic synthesis of organic molecules*** in a reducing atmosphere is possible.

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- However, the evidence is not yet convincing that the early atmosphere was in fact reducing.
 - Instead of forming in the atmosphere, the first organic compounds may have been synthesized near submerged volcanoes and deep-sea vents.
 - Amino acids have also been found in meteorites.

Deep Sea Vents



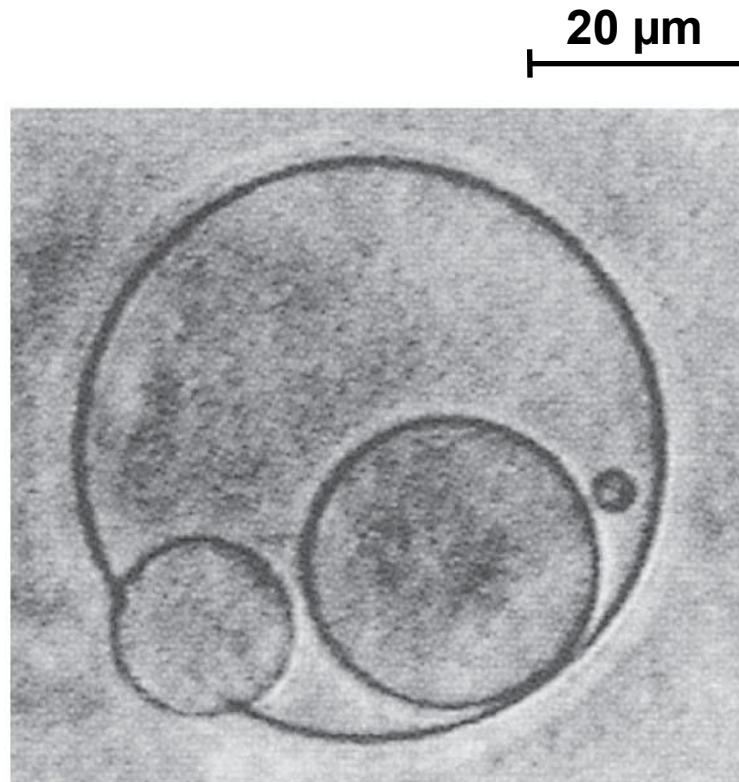
Abiotic Synthesis of Macromolecules

Monomers --> Polymers

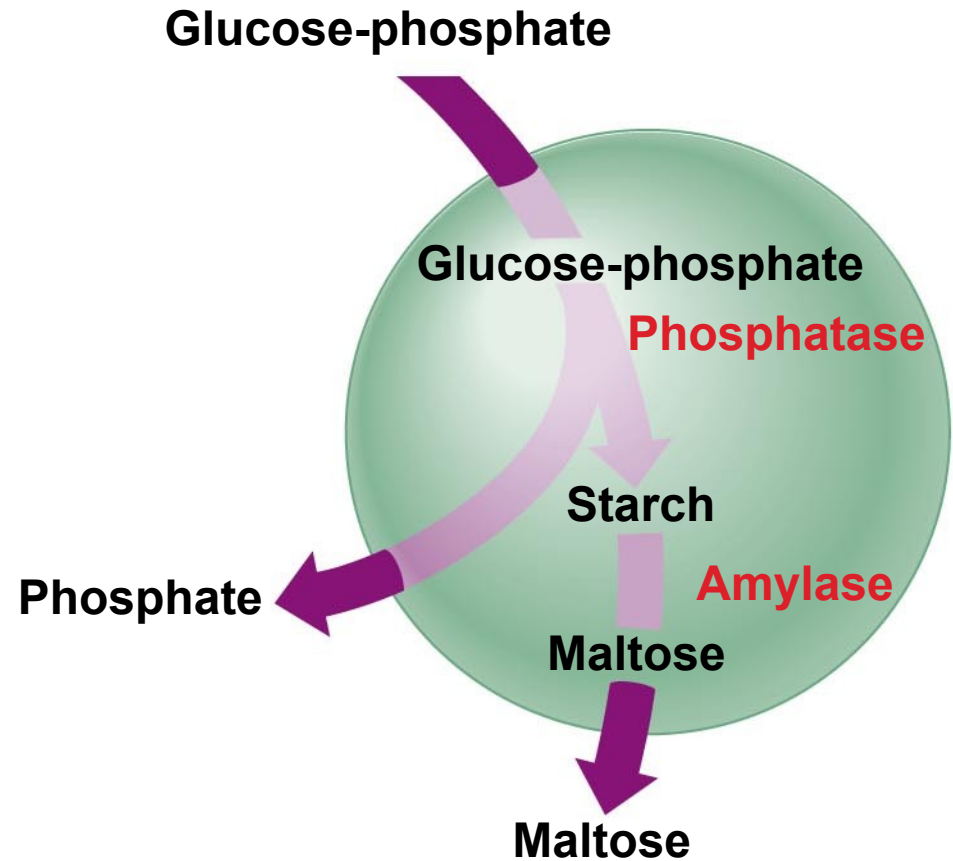
- Small organic molecules polymerize when they are concentrated on hot sand, clay, or rock.
- Replication and metabolism are key properties of life.
- **Protobionts** are *aggregates* of abiotically produced molecules *surrounded by a membrane* or membrane-like structure.
- Protobionts *exhibit simple reproduction and metabolism* and maintain an internal chemical environment.

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- Experiments demonstrate that protobionts could have formed spontaneously from abiotically produced organic compounds.
 - For example, small membrane-bounded droplets called liposomes can form when lipids or other organic molecules are added to water.

Protopionts May Have Formed Spontaneously



(a) Simple reproduction by liposomes



(b) Simple metabolism

Self-Replicating **RNA** and the Dawn of Natural Selection

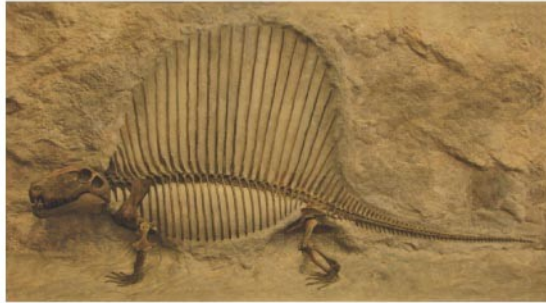
- The **first genetic material** was probably **RNA**, not DNA.
- **RNA molecules** called **ribozymes** have been found to **catalyze** many different **reactions**
 - For example, ribozymes can make complementary copies of short stretches of their own sequence or other short pieces of RNA.

Sedimentary Rocks and Fossils

- **Sedimentary strata** reveal the **relative ages** of fossils.
- The **absolute ages** of fossils can be determined by **radiometric dating**.
- A “parent” isotope decays to a “daughter” isotope at a constant rate.
- Each isotope has a known **half-life**, the time required for half the parent isotope to decay.

Sedimentary Rock Strata -- Fossils

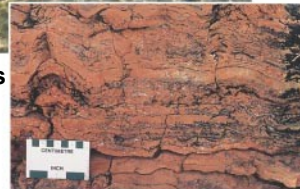
▼ *Dimetrodon*



▲ *Coccothecus cuspidatus*



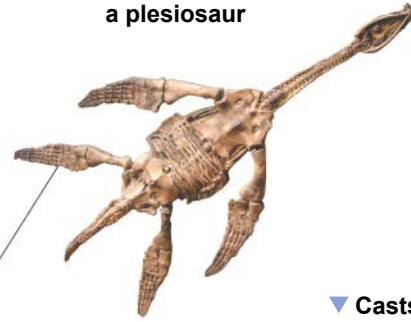
▲ Stromatolites



▲ Fossilized stromatolite

Present

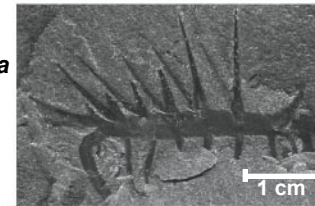
▼ *Rhomaleosaurus victor*, a plesiosaur



▼ Casts of ammonites



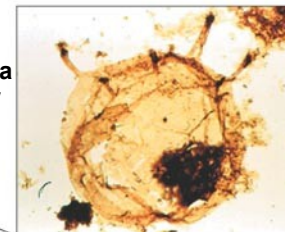
▶ *Hallucigenia*



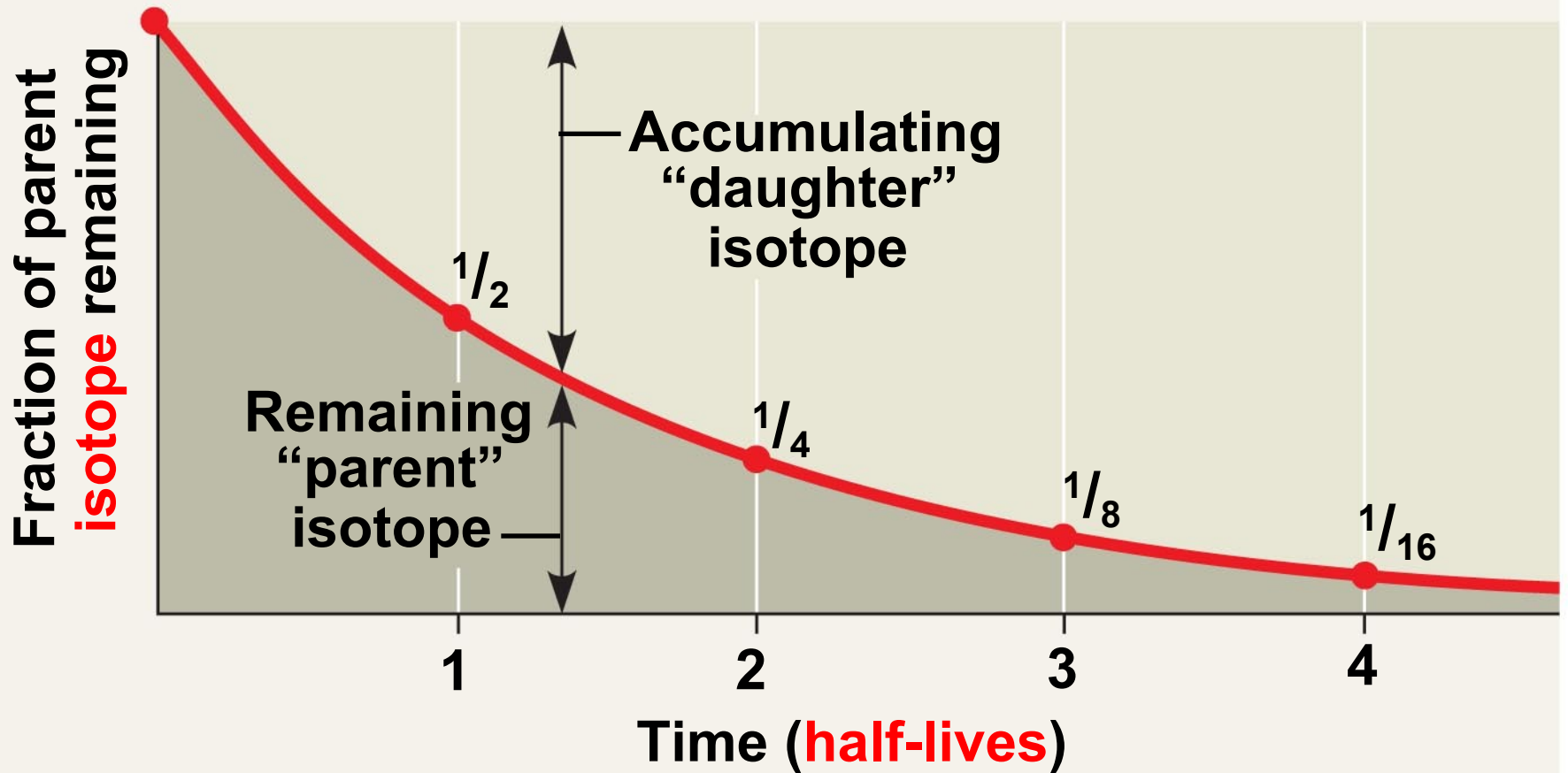
▶ *Dickinsonia costata*



▶ *Tappania*, a unicellular eukaryote



Radiometric Dating



-
- Radiocarbon dating can be used to date fossils up to 75,000 years old.
 - For older fossils, some isotopes can be used to date sedimentary rock layers above and below the fossil.
 - The magnetism of rocks can provide dating information.
 - Reversals of the magnetic poles leave their record on rocks throughout the world.

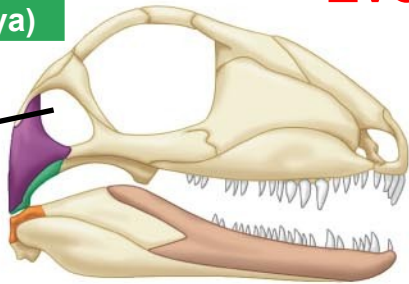
The Origin of New Groups of Organisms

- Mammals belong to the group of animals called *tetrapods*.
- The evolution of unique mammalian features through gradual modifications can be traced from ancestral synapsids through the present.

Evolution of Mammals

Synapsid (300 mya)

Temporal fenestra

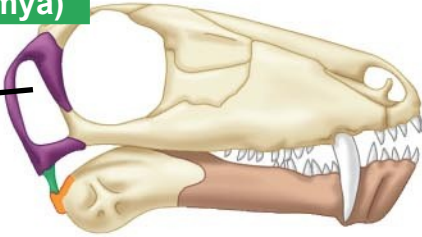


Key

Articular Dentary
Quadrate Squamosal

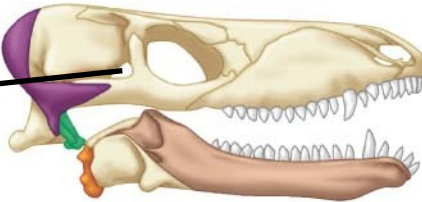
Therapsid (280 mya)

Temporal fenestra

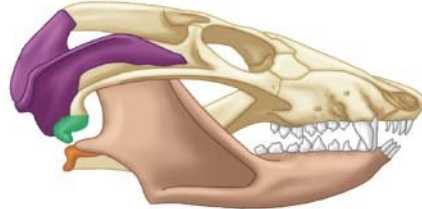


Early cynodont (260 mya)

Temporal fenestra



Later cynodont (220 mya)



Very late cynodont (195 mya)



EARLY
TETRAPODS

Synapsids

Therapsids

Earlier cynodonts

Reptiles
(including
dinosaurs and birds)



Dimetrodon



Very late cynodonts

Mammals



Concept 25.3: **Key events in life's history** include the origins of single-celled and multicelled organisms and the colonization of land

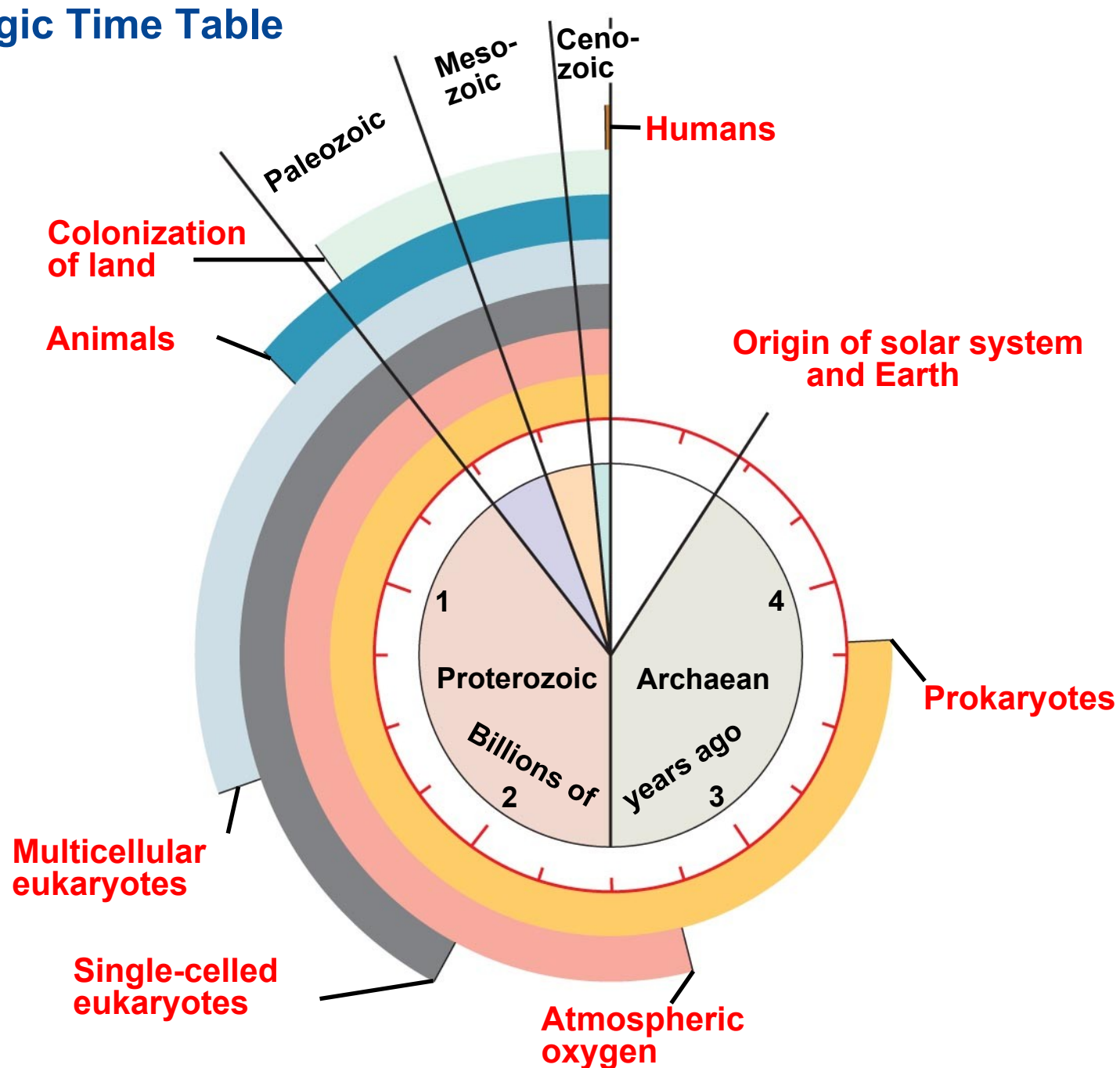
- The **geologic record** is divided into the Archaean, the Proterozoic, and the Phanerozoic eons.
- The Phanerozoic encompasses multicellular eukaryotic life.
- The Phanerozoic is divided into three eras: the Paleozoic, Mesozoic, and Cenozoic.
- Major boundaries between geological divisions correspond to extinction events in the fossil record.

Geologic Record

Table 25.1 The Geologic Record

Relative Duration of Eons	Era	Period	Epoch	Age (Millions of Years Ago)	Some Important Events in the History of Life
Phanerozoic	Cenozoic	Neogene	Holocene	0.01	Historical time
			Pleistocene	1.8	Ice ages; humans appear
			Pliocene	5.3	Origin of genus <i>Homo</i>
			Miocene	23	Continued radiation of mammals and angiosperms; apelike ancestors of humans appear
		Paleogene	Oligocene	33.9	Origins of many primate groups, including apes
			Eocene	55.8	Angiosperm dominance increases; continued radiation of most present-day mammalian orders
			Paleocene	65.5	Major radiation of mammals, birds, and pollinating insects
Proterozoic	Mesozoic	Cretaceous		145.5	Flowering plants (angiosperms) appear and diversify; many groups of organisms, including most dinosaurs, become extinct at end of period
		Jurassic		199.6	Gymnosperms continue as dominant plants; dinosaurs abundant and diverse
		Triassic		251	Cone-bearing plants (gymnosperms) dominate landscape; dinosaurs evolve and radiate; origin of mammals
	Paleozoic	Permian		299	Radiation of reptiles; origin of most present-day groups of insects; extinction of many marine and terrestrial organisms at end of period
		Carboniferous		359.2	Extensive forests of vascular plants form; first seed plants appear; origin of reptiles; amphibians dominant
		Devonian		416	Diversification of bony fishes; first tetrapods and insects appear
		Silurian		443.7	Diversification of early vascular plants
		Ordovician		488.3	Marine algae abundant; colonization of land by diverse fungi, plants, and animals
		Cambrian		542	Sudden increase in diversity of many animal phyla (Cambrian explosion)
Archaean	Ediacaran			635	Diverse algae and soft-bodied invertebrate animals appear
				2,100	Oldest fossils of eukaryotic cells appear
				2,500	Concentration of atmospheric oxygen begins to increase
				3,500	Oldest fossils of cells (prokaryotes) appear
				3,800	Oldest known rocks on Earth's surface
				Approx. 4,600	Origin of Earth

Geologic Time Table



The First Single-Celled Organisms = **Prokaryotes**

- The **oldest** known **fossils** are **stromatolites**, rock-like structures composed of many layers of **bacteria** and sediment.
- Stromatolites date back 3.5 billion years ago
- Prokaryotes were Earth's sole inhabitants from 3.5 to about 2.1 billion years ago.

Photosynthesis and the Oxygen Revolution

- Most atmospheric oxygen (O_2) is of biological origin.
- O_2 produced by oxygenic **photosynthesis** reacted with dissolved iron and precipitated out to form banded iron formations.
- The **source of O_2** was likely bacteria similar to modern **cyanobacteria**.

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- By about 2.7 billion years ago, O₂ began accumulating in the atmosphere and rusting iron-rich terrestrial rocks.
 - This “*oxygen revolution*” from 2.7 to 2.2 billion years ago
 - Posed a challenge for life
 - Provided opportunity to gain energy from light
 - *Allowed organisms to exploit new ecosystems.*

About 2.7 billion years ago, O_2 began accumulating in the atmosphere and rusting iron-rich terrestrial rocks.

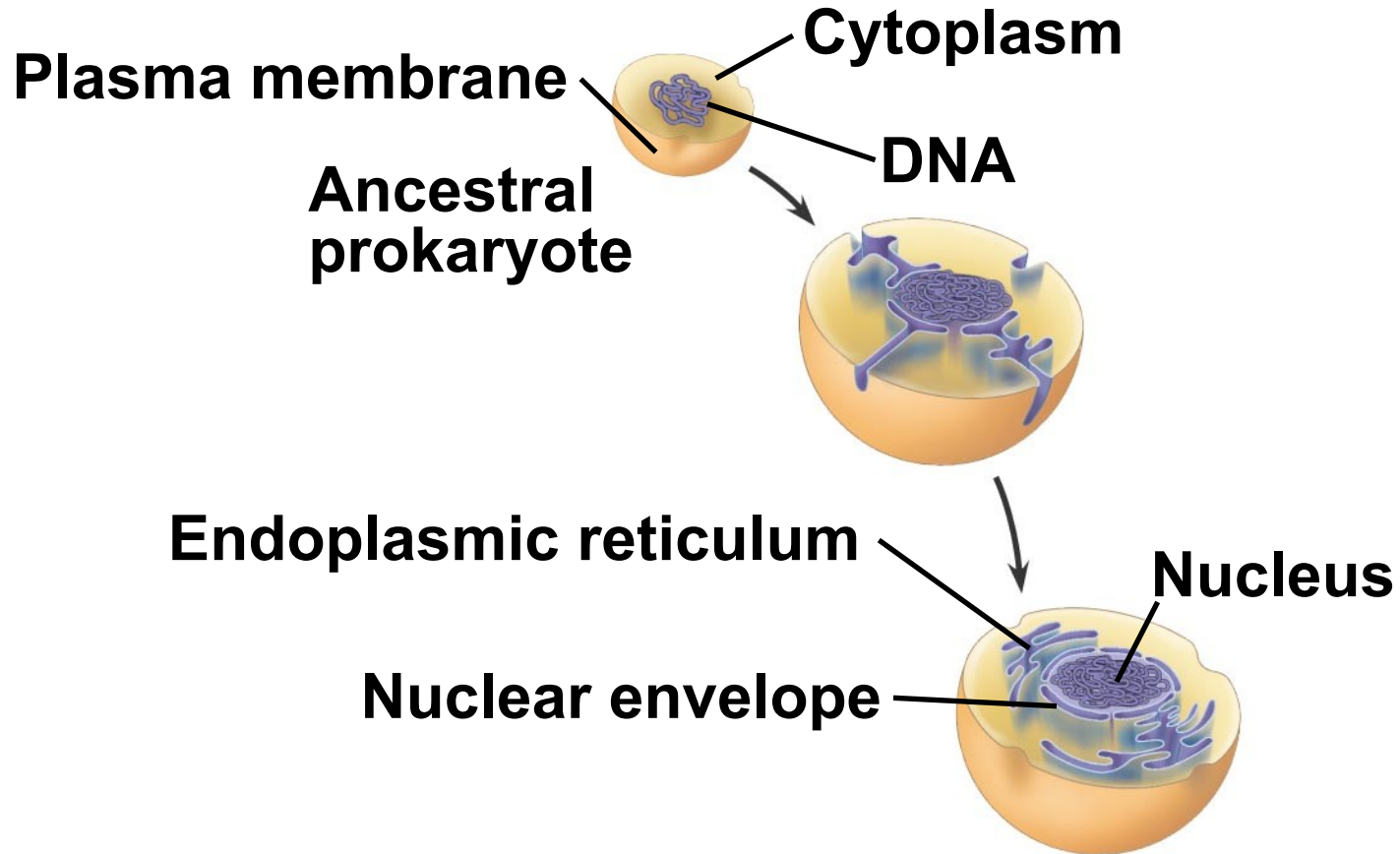


The First Eukaryotes

- The oldest fossils of eukaryotic cells date back 2.1 billion years.
- The hypothesis of **endosymbiosis** proposes that **mitochondria** and **plastids** (chloroplasts and related organelles) were **formerly** small **prokaryotes** living within larger host cells
- An **endosymbiont** is a *cell that lives within a host cell*.

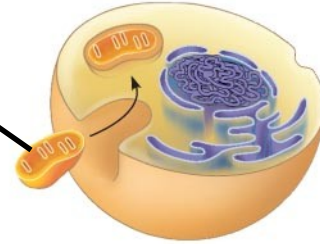
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- The prokaryotic ancestors of mitochondria and plastids probably gained entry to the host cell as undigested prey or internal parasites.
 - In the process of becoming more interdependent, the host and endosymbionts would have become a single organism.
 - ***Serial endosymbiosis*** supposes that mitochondria evolved before plastids through ***a sequence of endosymbiotic events***.

Invagination of Plasma Membrane



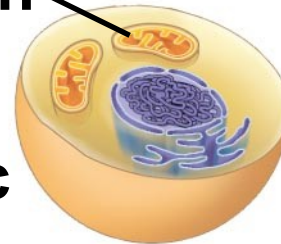
Serial Endosymbiosis

**Aerobic
heterotrophic
prokaryote**



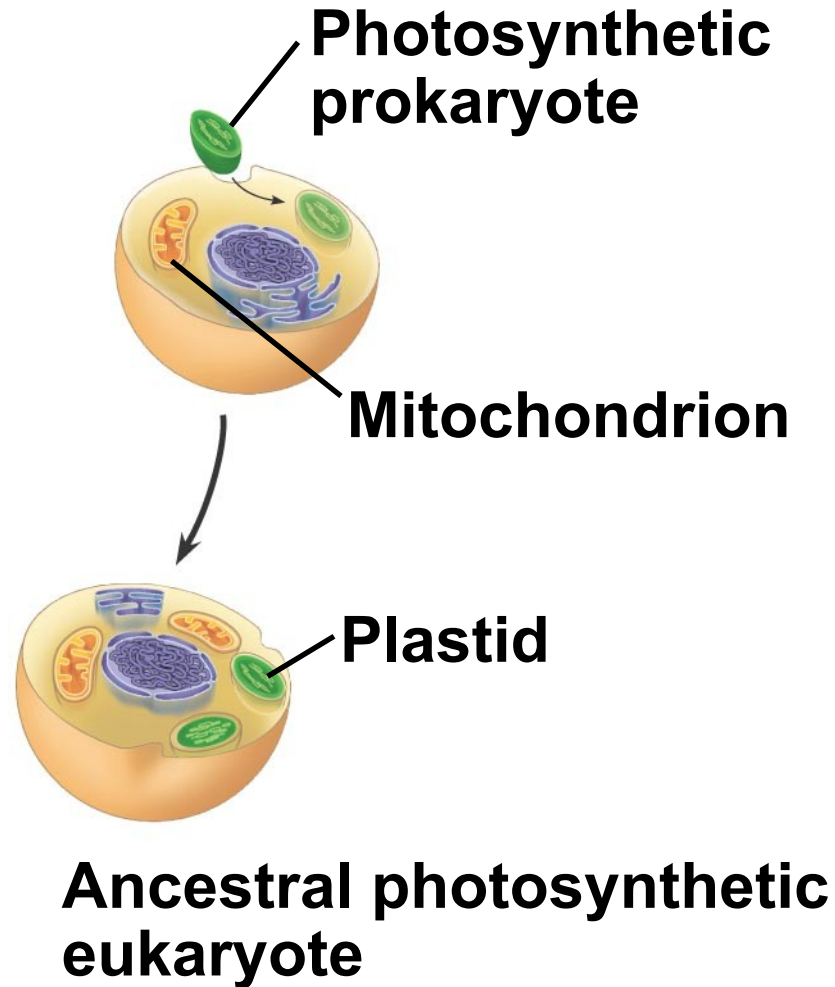
Mitochondrion

**Ancestral
heterotrophic
eukaryote**

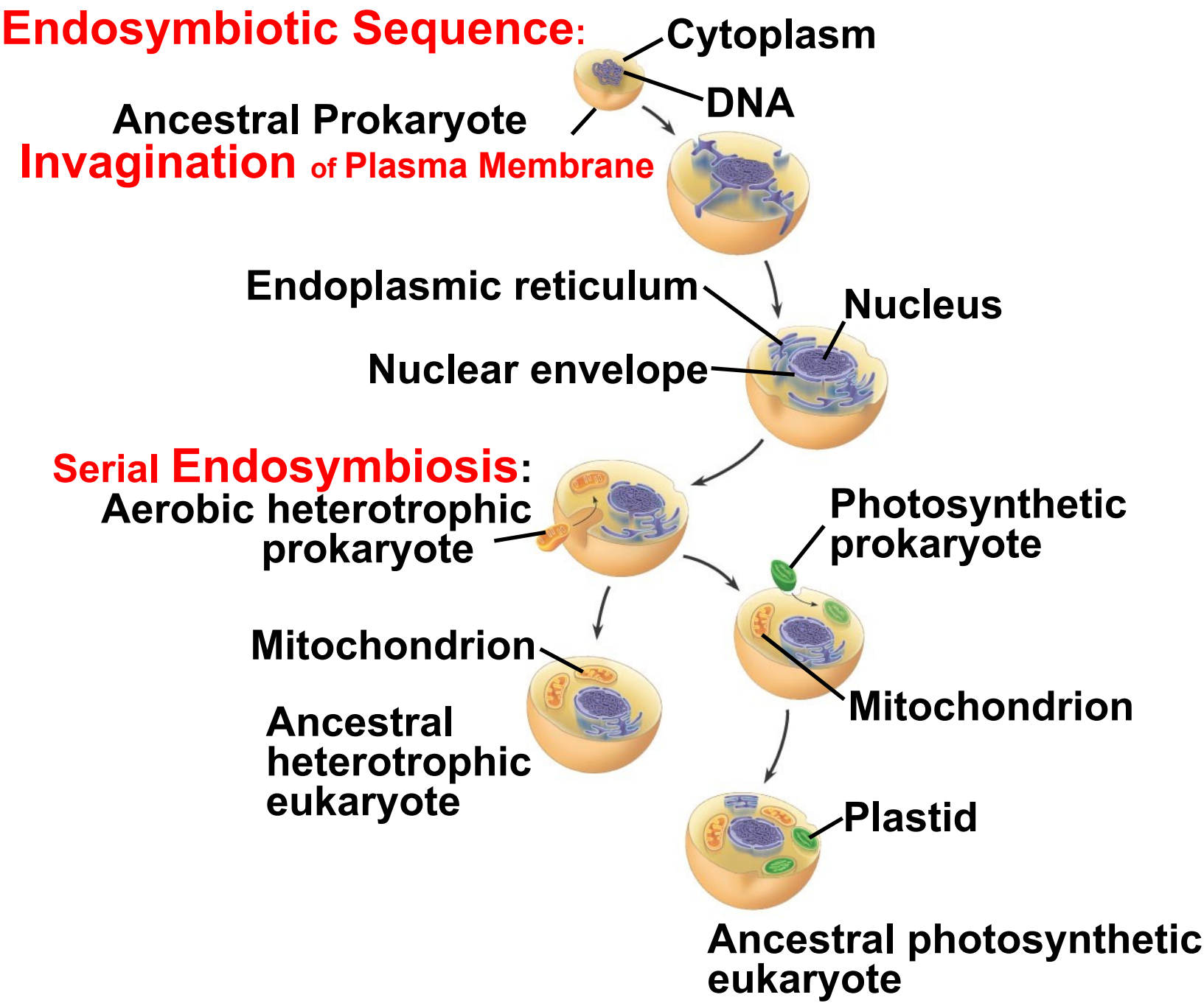


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Serial Endosymbiosis



Endosymbiotic Sequence:



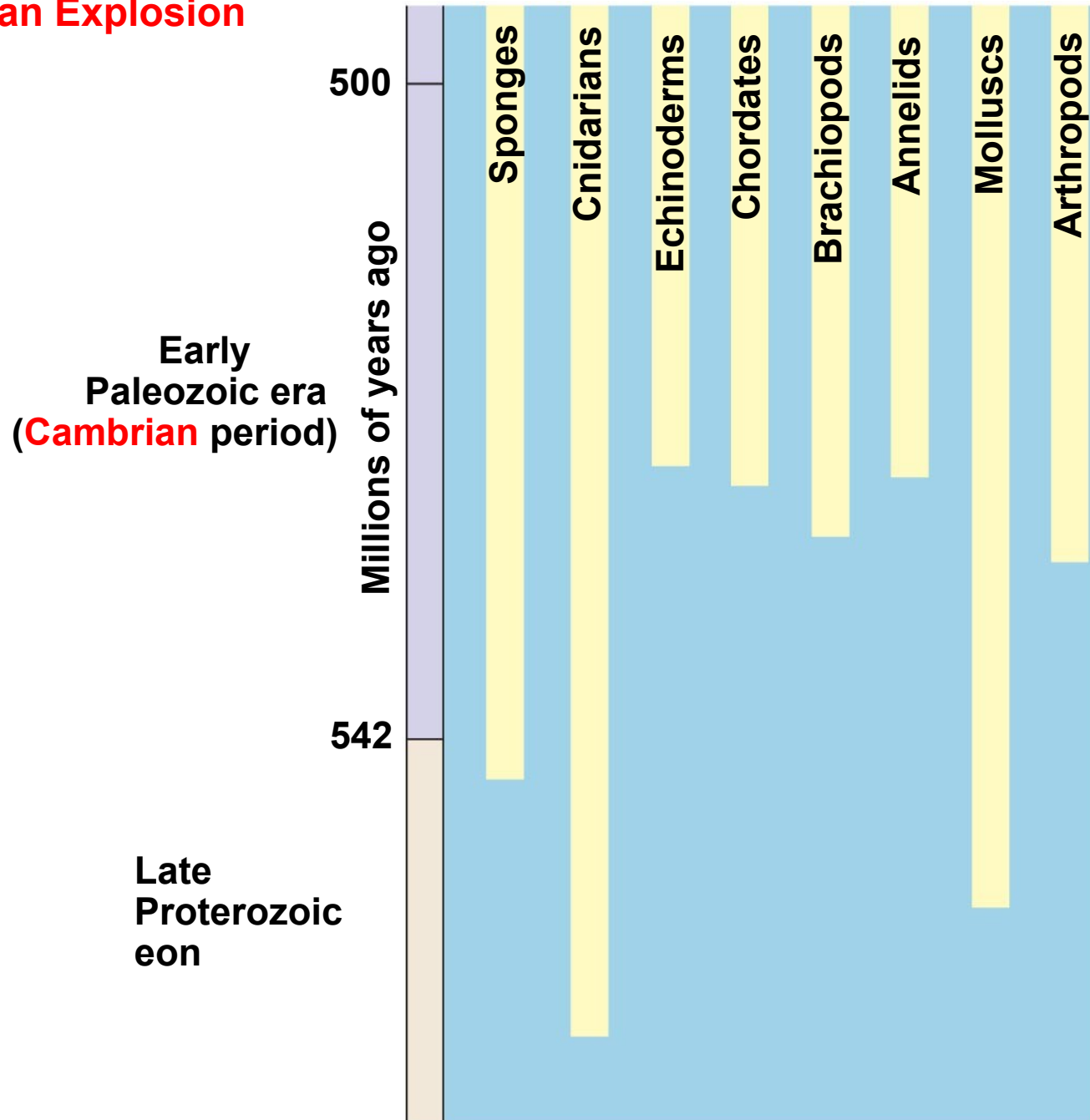
-
- Key **evidence** supporting an **endosymbiotic origin** of **mitochondria** and **plastids**:
 - Similarities in inner membrane structures and functions.
 - These organelles transcribe and translate their **own DNA**.
 - Their **ribosomes** are more similar to **prokaryotic** than eukaryotic ribosomes.

The Origin of Multicellularity

- The evolution of eukaryotic cells allowed for a greater range of unicellular forms.
- A second wave of diversification occurred when multicellularity evolved and gave rise to algae, plants, fungi, and animals.
- Comparisons of DNA sequences date the common ancestor of multicellular eukaryotes to 1.5 billion years ago.
- The **oldest known fossils of multicellular eukaryotes** are of **small algae** that lived about 1.2 billion years ago.

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- The “*snowball Earth*” *hypothesis* suggests that periods of **extreme glaciation** confined life to the equatorial region or deep-sea vents from 750 to 580 million years ago.
 - The ***Cambrian explosion*** refers to the *sudden appearance of fossils resembling modern phyla* in the Cambrian period (535 to 525 million years ago).
 - The Cambrian explosion provides the first evidence of *predator-prey interactions*.
 - Fossils in China provide evidence of modern animal phyla tens of millions of years before the Cambrian explosion.

Cambrian Explosion



Proterozoic Fossils that may be animal embryos (SEM)



(a) Two-cell stage

150 μm



(b) Later stage

200 μm

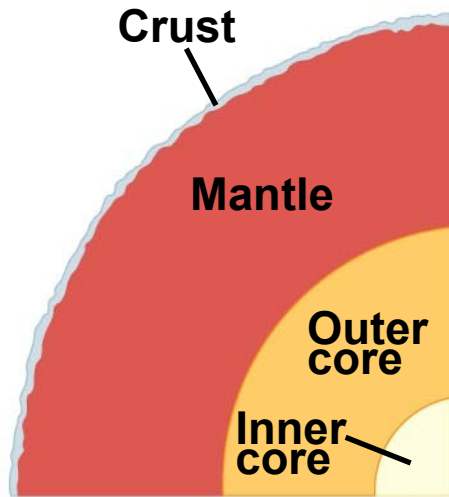
The Colonization of Land

- Fungi, plants, and animals began to colonize land about **500 million years ago**.
- Plants and fungi likely colonized land together by 420 million years ago.
- Arthropods and tetrapods are the most widespread and diverse land animals.
- Tetrapods evolved from lobe-finned fishes around 365 million years ago.

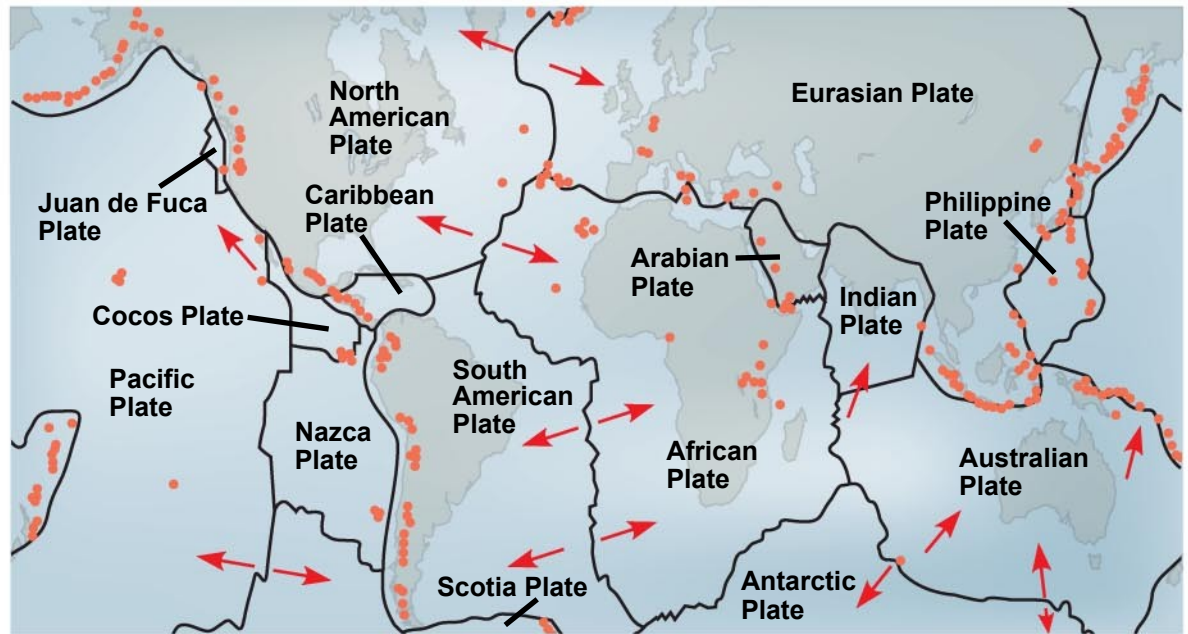
Concept 25.4: The rise and fall of dominant groups reflect continental drift, mass extinctions, and adaptive radiations

- At three points in time, the land masses of Earth have formed a supercontinent: 1.1 billion, 600 million, and 250 million years ago.
- Earth's continents move slowly over the underlying hot mantle through the process of **continental drift**.
- Oceanic and continental plates can collide, separate, or slide past each other.
- Interactions between plates cause the formation of mountains and islands, and earthquakes.

Earth - Plate Tectonics: Continental Drift



(a) Cutaway view of Earth

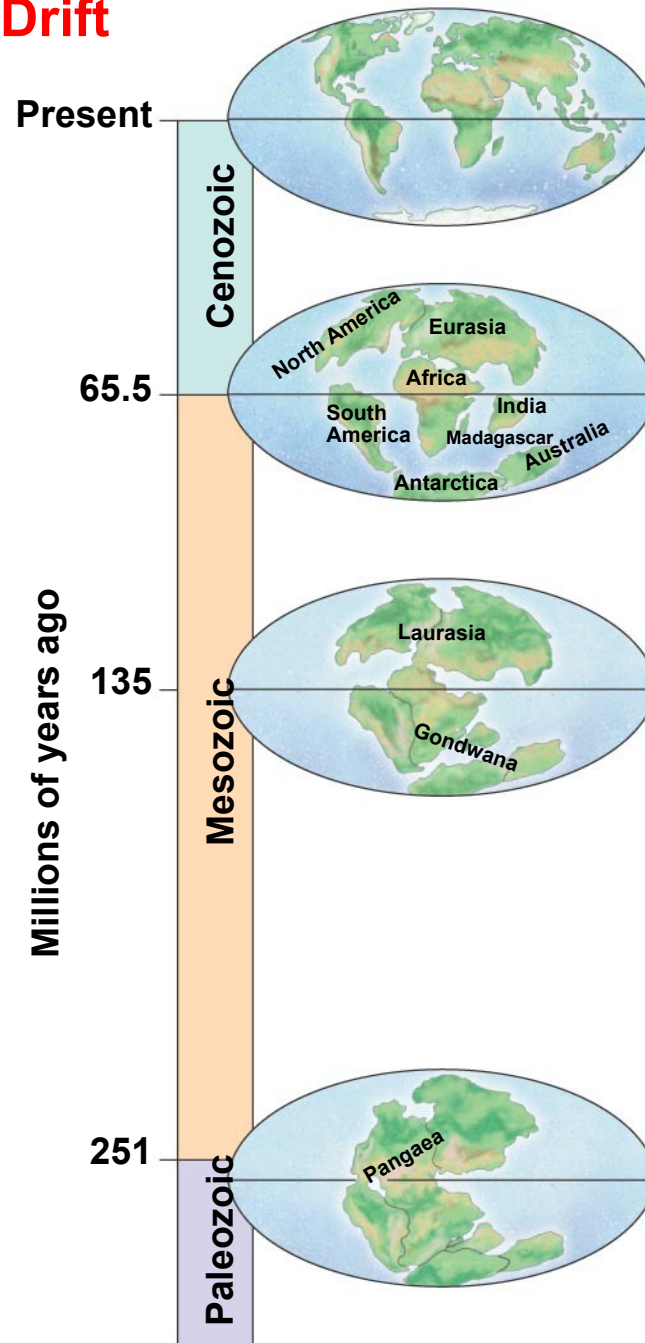


(b) Major continental plates

Consequences of Continental Drift

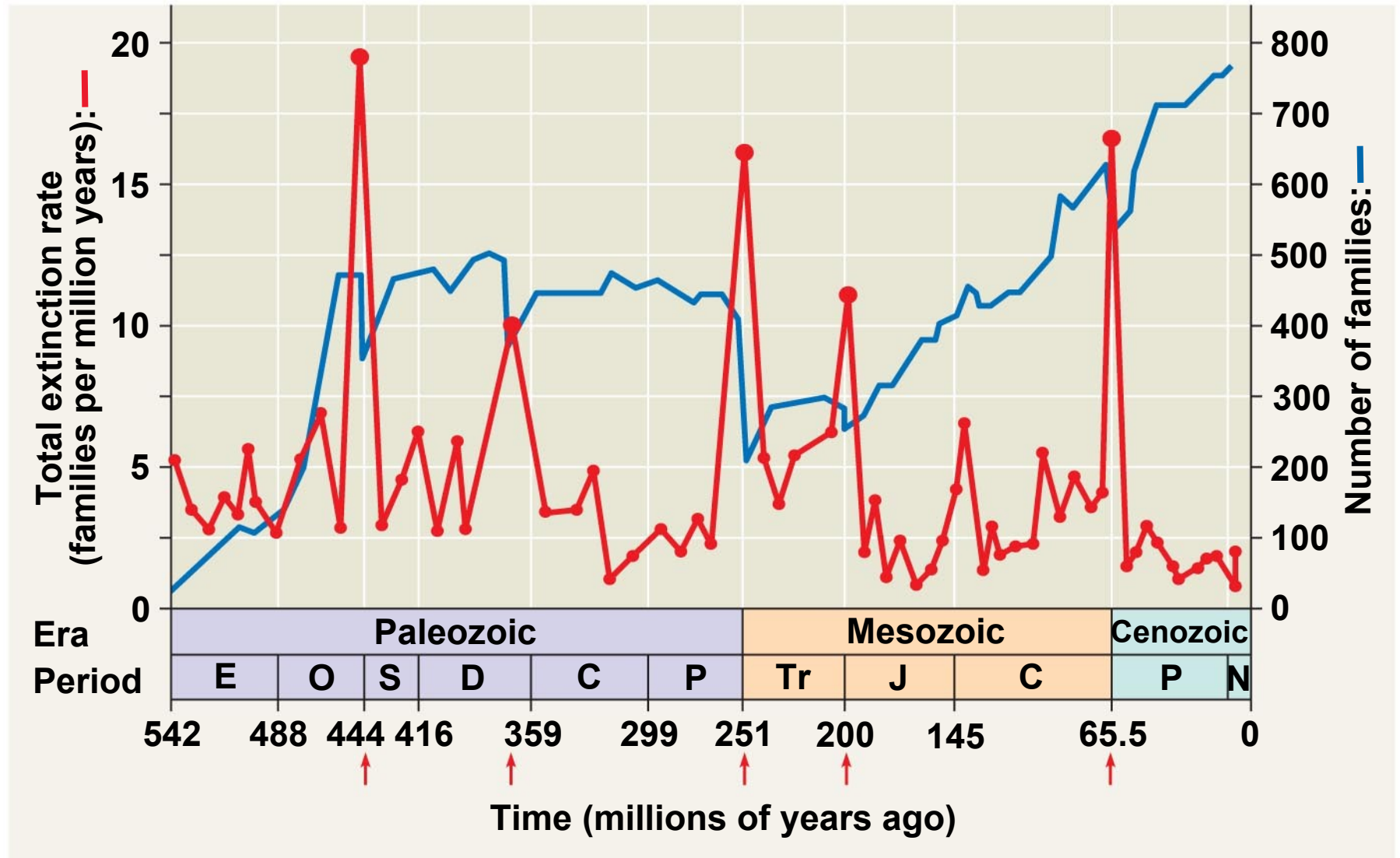
- Formation of the supercontinent **Pangaea** about 250 million years ago had many effects:
 - A reduction in shallow water habitat
 - A colder and drier climate inland
 - Changes in climate as continents moved toward and away from the poles
 - Changes in ocean circulation patterns leading to global cooling.

History of Continental Drift



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- The break-up of Pangaea lead to **allopatric speciation**.
 - The current distribution of fossils reflects the movement of **continental drift**. Similarity of fossils in parts of South America and Africa supports the idea that these continents were formerly attached.
 - The **fossil record** shows that most species that have ever lived are now extinct.
 - At times, the rate of extinction has increased dramatically and caused a **mass extinction**.
 - In each of the five mass extinction events, more than 50% of Earth's species became extinct.

Five Big Mass Extinctions

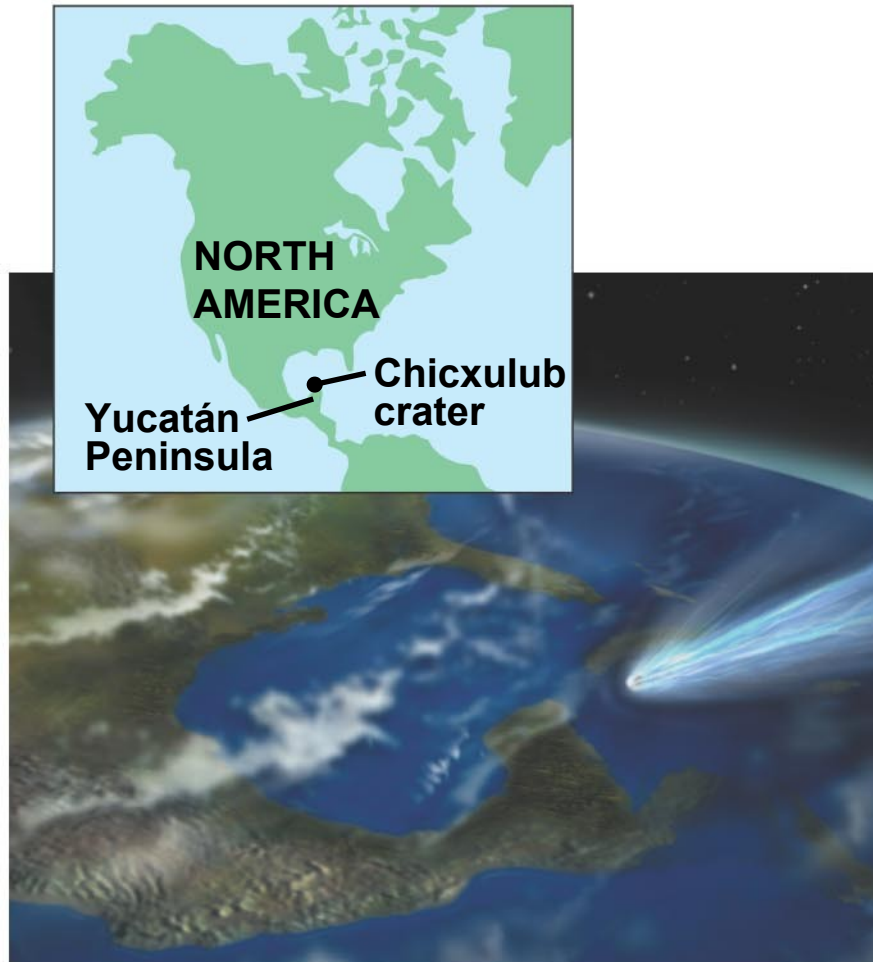


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- The Permian extinction defines the boundary between the Paleozoic and Mesozoic eras.
 - This mass extinction caused the extinction of about 96% of marine animal species and might have been caused by volcanism, which lead to global warming, and a decrease in oceanic oxygen.
 - The Cretaceous mass extinction 65.5 million years ago separates the Mesozoic from the Cenozoic.
 - Organisms that went extinct include about half of all marine species and many terrestrial plants and animals, including most dinosaurs.

Massive Meteorite Impact Evidence

- The presence of iridium in sedimentary rocks suggests a meteorite impact **about 65 million years ago**.
- The Chicxulub crater off the coast of Mexico is evidence of a meteorite that dates to the same time.

Evidence of Meteorite Impact



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Is a Sixth Mass Extinction Under Way?

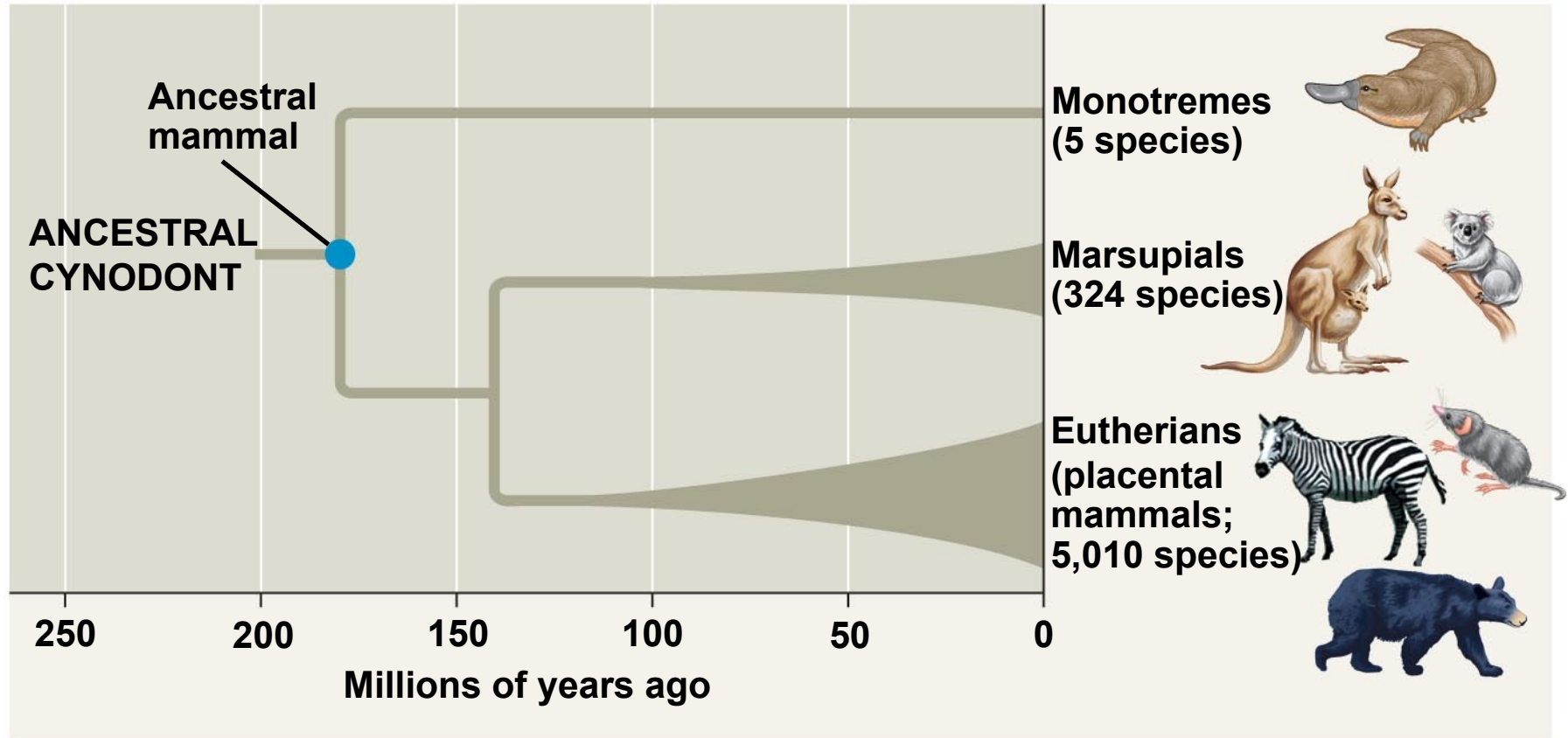
Consequences ...

- Scientists estimate that the current rate of extinction is 100 to 1,000 times the typical background rate.
- Data suggest that a sixth human-caused mass extinction is likely to occur unless dramatic action is taken.
- **Mass extinction** can **alter ecological communities** and the niches available to organisms.
- It can take from 5 to 100 million years for diversity to recover following a mass extinction.
- Mass extinction can pave the way for **adaptive radiations**.

Adaptive Radiations - New Environmental Opportunities ...

- ***Adaptive radiation*** is the evolution of diversely adapted species from a common ancestor upon introduction to *new environmental opportunities*.
- Mammals underwent an adaptive radiation after the extinction of terrestrial dinosaurs.
- The disappearance of dinosaurs (except birds) allowed for the expansion of mammals in diversity and size.
- Other notable radiations include photosynthetic prokaryotes, large predators in the Cambrian, land plants, insects, and tetrapods.

World-Wide Adaptive Radiations



Regional Adaptive Radiations

- Adaptive radiations can occur when organisms colonize new environments with little competition.
- The Hawaiian Islands are one of the world's great showcases of adaptive radiation.

Hawaiian Islands -- Regional Adaptive Radiations



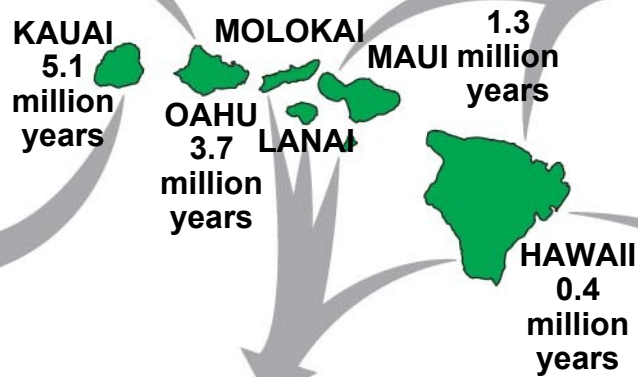
Dubautia laxa



Close North American relative,
the tarweed *Carlquistia muirii*



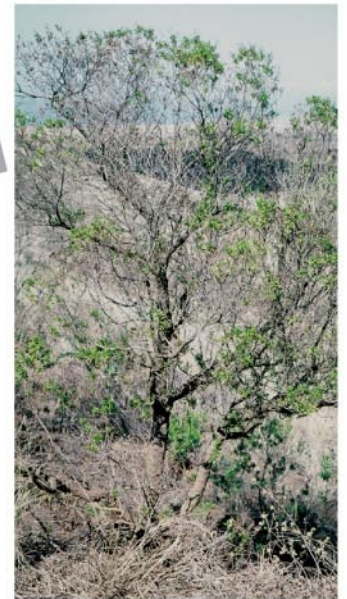
Argyroxiphium sandwicense



Dubautia waialealae



Dubautia scabra

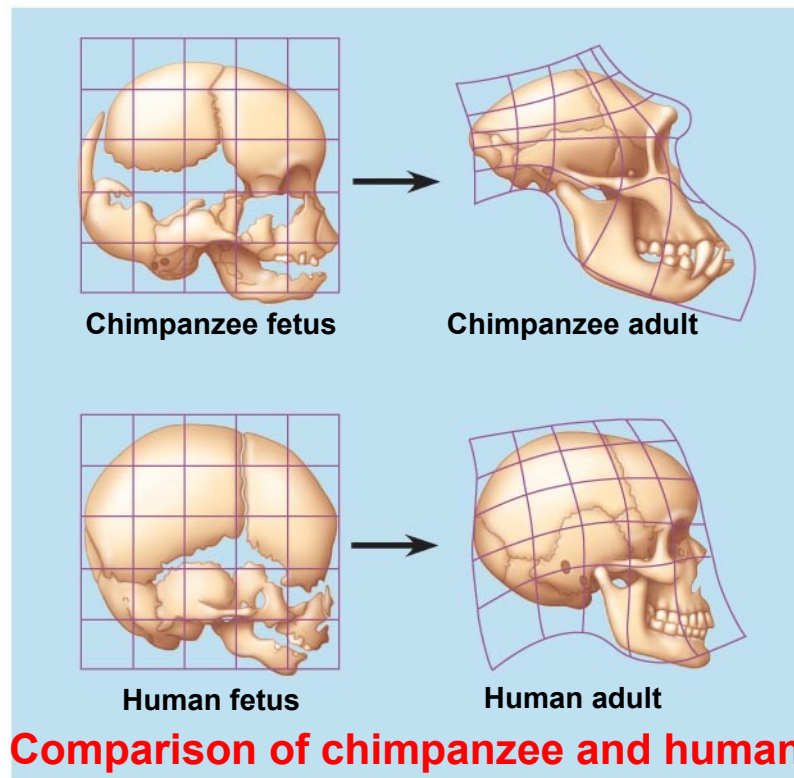
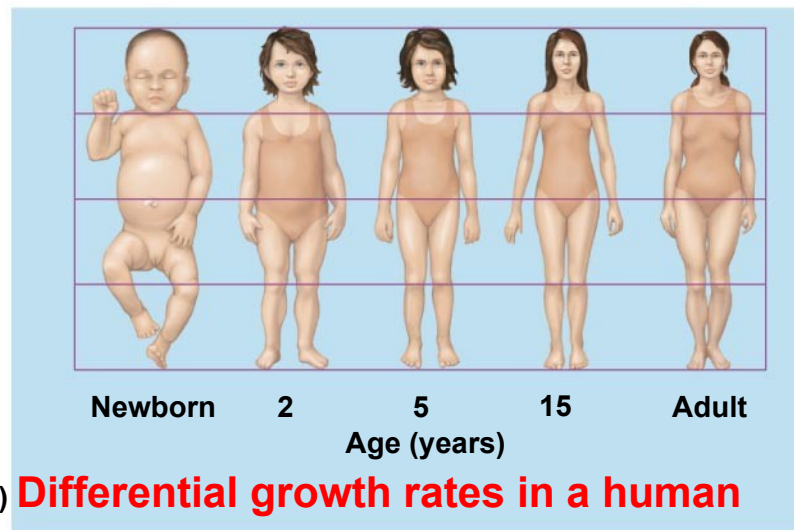


Dubautia linearis

Major changes in body form result from changes in the sequences and regulation of developmental genes

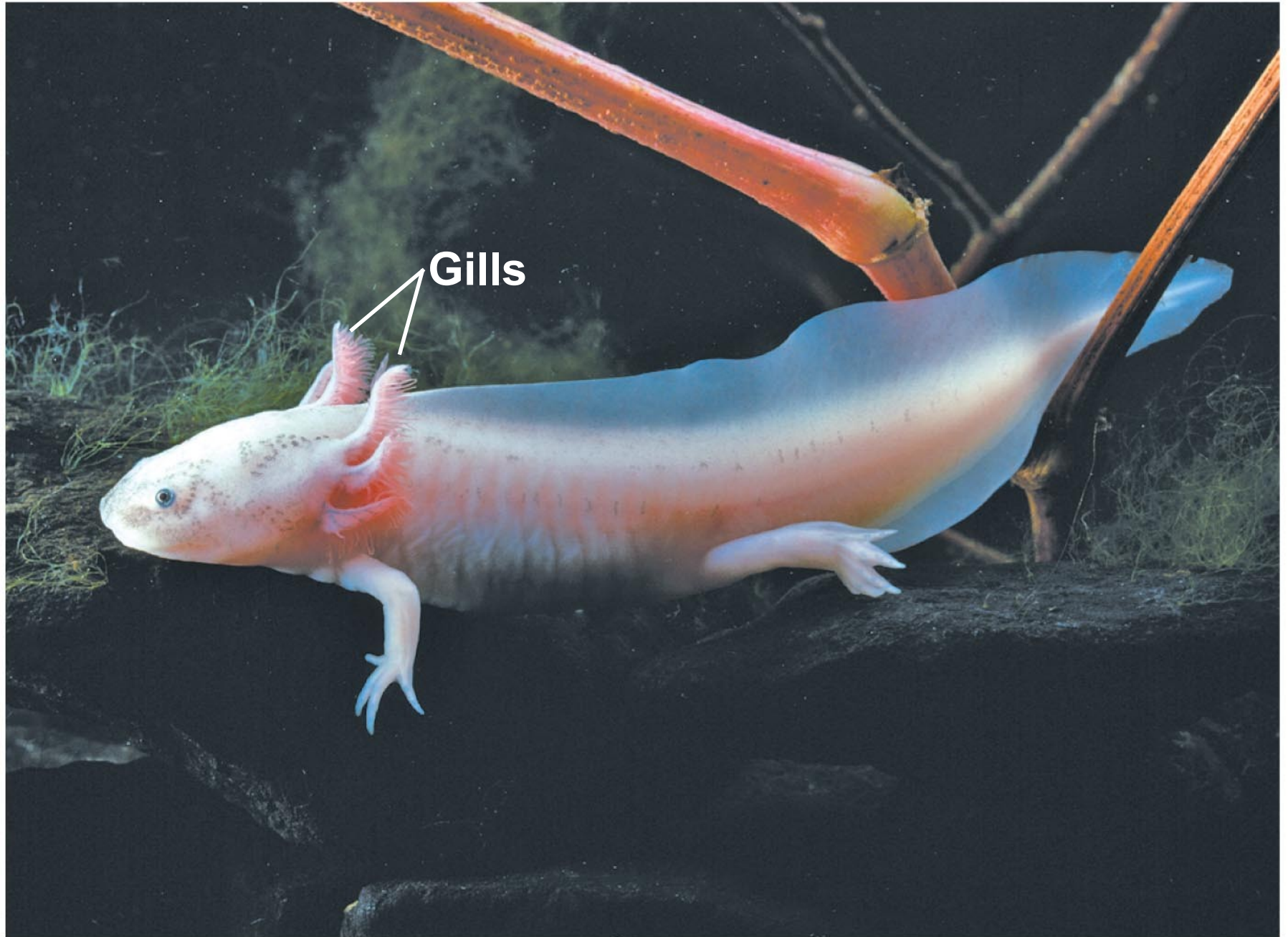
- Studying genetic mechanisms of change can provide insight into large-scale evolutionary change.
- Genes that program development control the rate, timing, and spatial pattern of changes in an organism's form as it develops into an adult.
- ***Heterochrony*** is an *evolutionary change in the rate or timing of developmental events*.
- It can have a significant impact on body shape.
- The contrasting shapes of human and chimpanzee skulls are the result of small changes in relative growth rates.

Allometric Growth



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- **Heterochrony** can alter the timing of reproductive development relative to the development of nonreproductive organs
 - In **paedomorphosis**, the rate of reproductive development accelerates compared with somatic development.
 - The sexually **mature** species **may retain body features that were juvenile** structures in an ancestral species.

Paedomorphosis - Juvenile Gills Retained by Adult Salamander



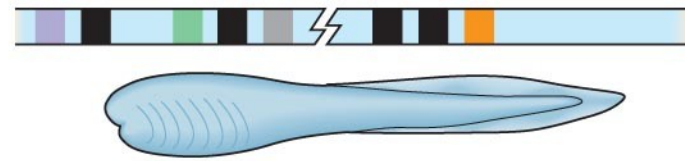
Changes in Spatial Pattern - Hox genes

- Substantial evolutionary change can also result from alterations in genes that control the placement and **organization of body parts**.
- **Homeotic genes** determine such basic features as where wings and legs will develop on a bird or how a flower's parts are arranged.
- **Hox genes** are a class of **homeotic genes** that provide **positional information during development**.
- If *Hox* genes are expressed in the wrong location, body parts can be produced in the wrong location.
- For example, in crustaceans, a swimming appendage can be produced instead of a feeding appendage.

-
- **Evolution of vertebrates** from invertebrate animals was associated with **alterations** in ***Hox genes***.
 - Two duplications of *Hox* genes have occurred in the vertebrate lineage.
 - These duplications may have been important in the evolution of new vertebrate characteristics.
 - The tremendous increase in diversity during the Cambrian explosion is a puzzle.
 - *Changes in developmental genes can also result in new morphological forms.*

Hox Genes Alterations

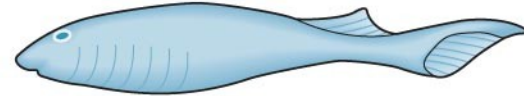
Hypothetical vertebrate ancestor (invertebrate) with a single *Hox* cluster



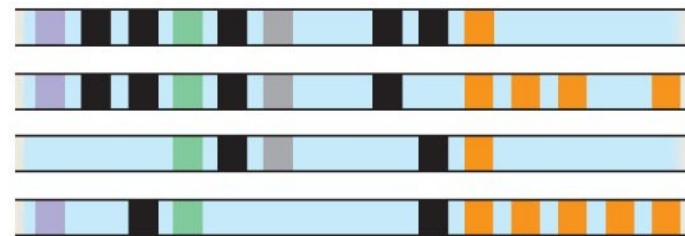
First *Hox* duplication



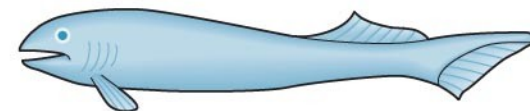
Hypothetical early vertebrates (jawless) with two *Hox* clusters



Second *Hox* duplication



Vertebrates (with jaws) with four *Hox* clusters



Changes in developmental genes can result in new morphological forms

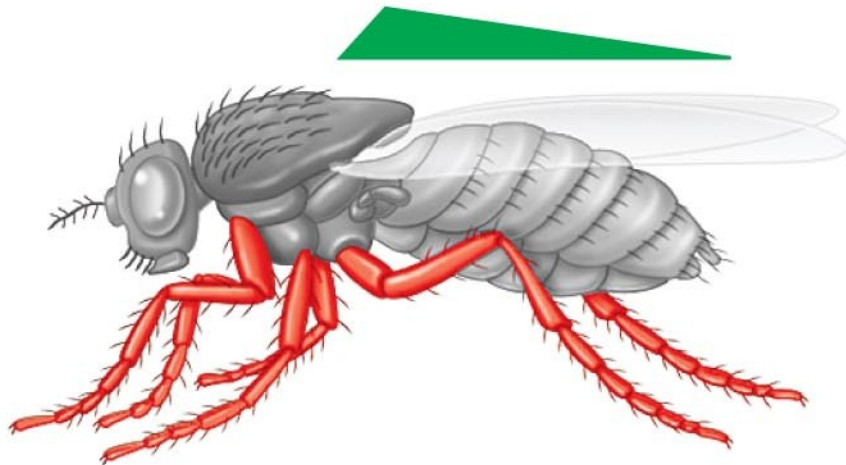
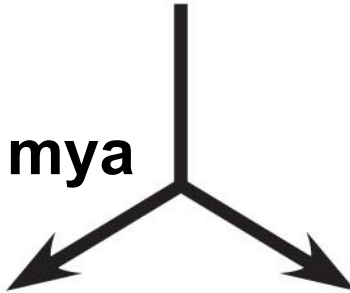
Hox gene 6

Hox gene 7

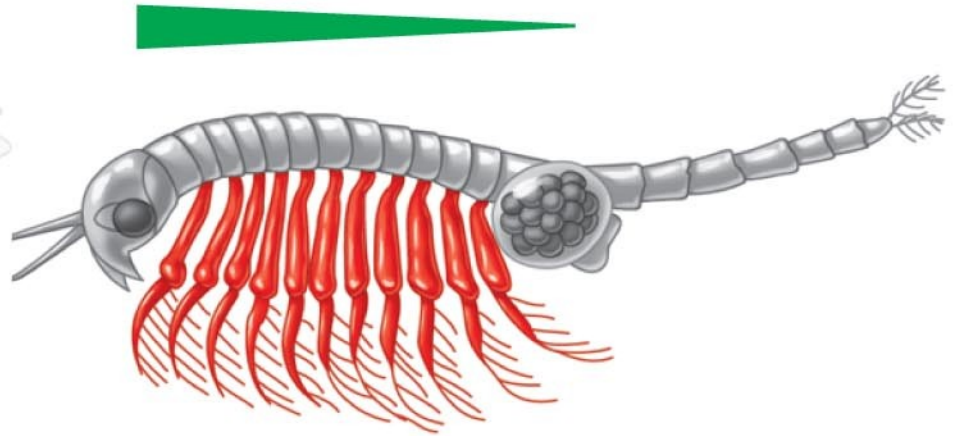
Hox gene 8



About 400 mya



Drosophila

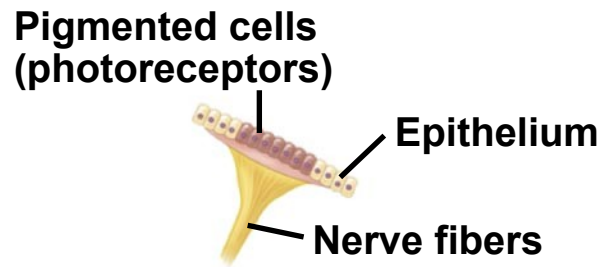


Artemia

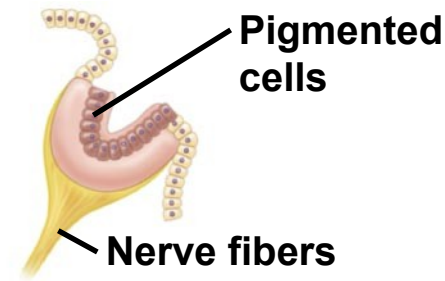
Concept 25.6: Evolution is not goal oriented

- Evolution is like tinkering—it is a process in which new forms arise by the slight modification of existing forms.
- *Most novel biological structures evolve in many stages from previously existing structures.*
- Complex eyes have evolved from simple photosensitive cells independently many times.
- Exaptations are structures that evolve in one context but become co-opted for a different function.
- Natural selection can only improve a structure in the context of its current utility.

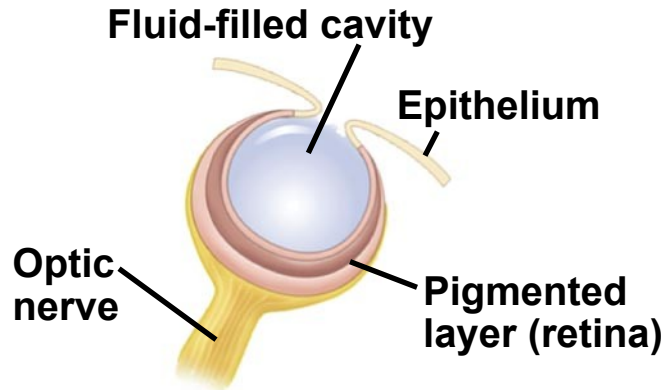
Evolution:
new forms
arise by the
slight
modification
of existing
forms



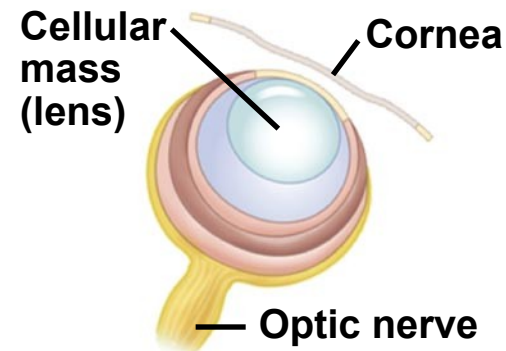
(a) Patch of pigmented cells



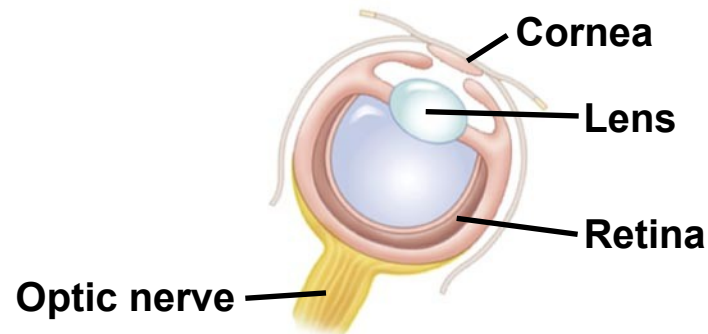
(b) Eyecup



(c) Pinhole camera-type eye

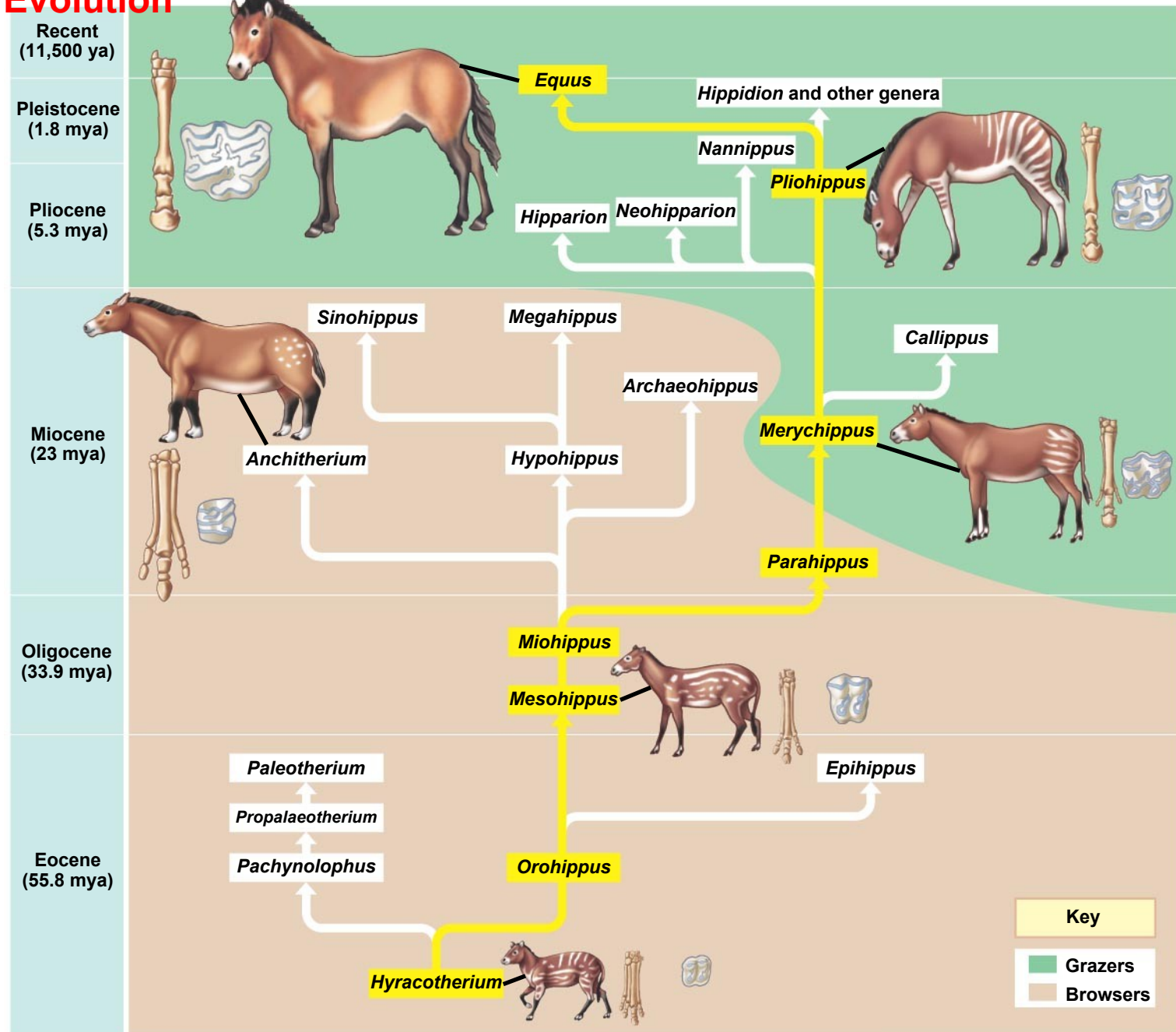


(d) Eye with primitive lens

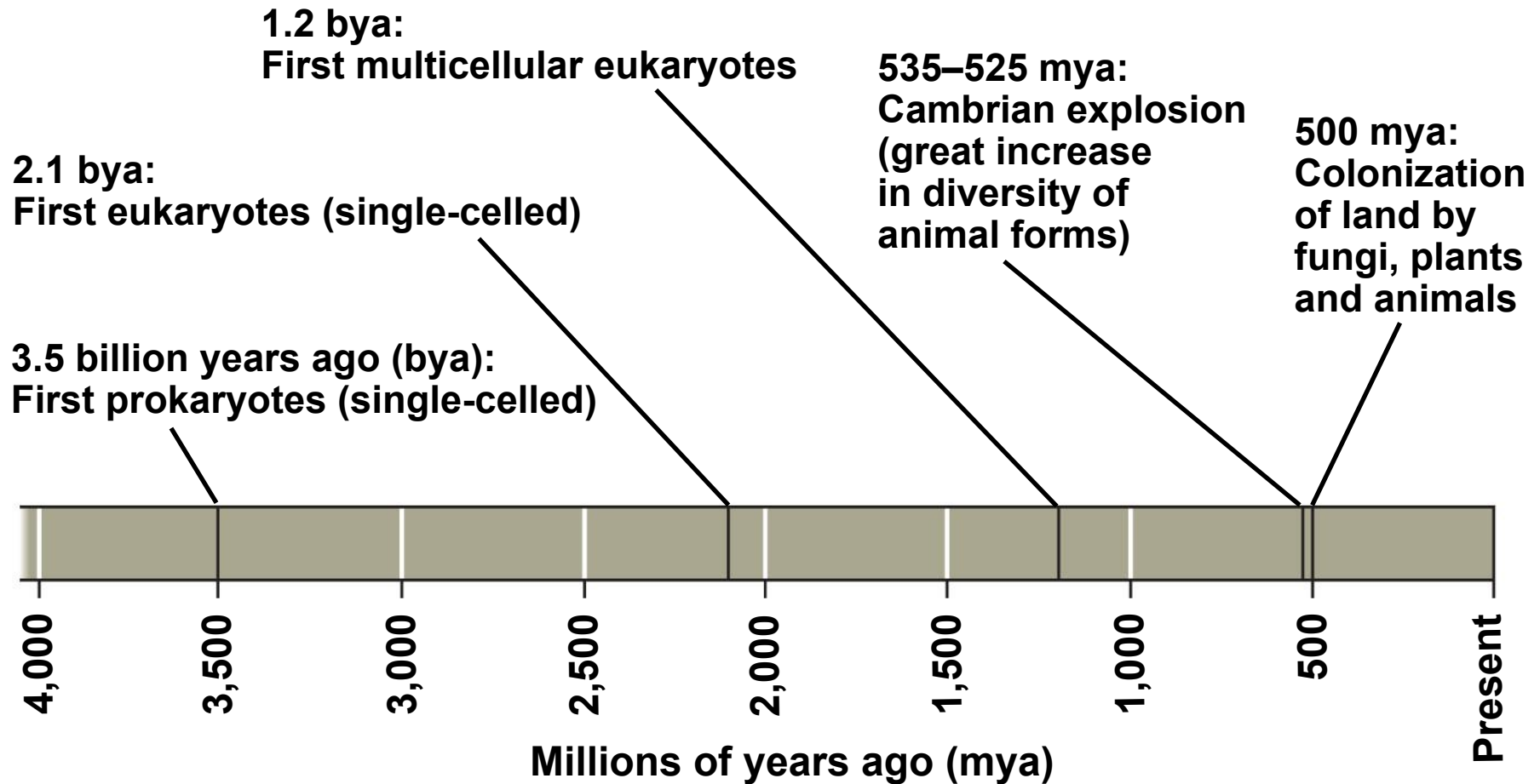


(e) Complex camera-type eye

Horse Evolution



The appearance of an evolutionary trend does not imply that there is some intrinsic drive toward a particular phenotype



You should now be able to:

1. Define radiometric dating, serial endosymbiosis, Pangaea, snowball Earth, exaptation, heterochrony, and paedomorphosis.
2. Describe the contributions made by Oparin, Haldane, Miller, and Urey toward understanding the origin of organic molecules.
3. Explain why RNA, not DNA, was likely the first genetic material.

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4. Describe and suggest evidence for the major events in the history of life on Earth from Earth's origin to 2 billion years ago.
 5. Briefly describe the Cambrian explosion.
 6. Explain how continental drift led to Australia's unique flora and fauna.
 7. Describe the mass extinctions that ended the Permian and Cretaceous periods.
 8. Explain the function of *Hox* genes.
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